

Nicholas A. Huang

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science (threads: Devices / Info-networks)

Atlanta, GA

August 2016 - May 2020

SKILLS

Software: Expert: Java, C#, Git, Linux/Unix Systems, Virtual Box, VMware, Docker, JetBrains Rider, Entity Framework

Intermediate in: SQL, ASP.Net Application Programming, Nmap, Wireshark, Kali Linux, Excel, Python, Kafka, System Design

Experienced in: Metasploit, Burp Suite, Cybersecurity Concepts

Language: Fluent English, Conversational Mandarin

Certifications

Oracle Generative AI Professional, AWS Cloud Practitioner, GCP Associate Cloud Engineer

WORK EXPERIENCE

Fortna Atlanta, GA

Software Engineer

February 2021 - November 2024

- Designed, programmed, and presented MHS's interactive robotics demonstration at ModeX; the demonstration was implemented using Node-Red, JavaScript, Mujin's robotic packaging systems, and Hai's robotic tote handling systems
- Implemented, tested, and deployed a WCS system for Advanced Auto Parts, providing package analysis, sortation logic, and exception handling
- Designed and implemented WCS functionality that allows for the export and import of a Json file for initial configuration of the system (i.e. starting scanner/container Id's, relations, configurations, etc.) using ASP.NET and Blazor
- Designed, implemented, and tested database models and DTOs (data transfer objects) to ensure that all data could be stored in a human readable fashion, retrieved efficiently, and transferred to objects that would be shared by the entire system
- Designed, implemented, and tested round robin and first full priority algorithms to enable to destination decision for the WCS system
- Discussed and contributed to the writing of proposals for purchasing equipment (cameras, AGV parts, server CPUs, etc.)
- Designed and developed MHS's first WCS system as a service called Helix; this not only allowed streamlining and faster software delivery to clients but saved the company time and resources allocated to projects
- Implemented the functionality and razor pages for PANDA (print and apply) in ASP.NET and Blazor; this allowed the user to be able to connect to printers, monitor printer statuses, change individual printer settings, and send ZPL data to test the format of labels
- Coordinated, designed, and programmed a message utility service (ASP.NET/C#) and razor page in the WCS system, giving system integrators the ability to send, monitor, and debug events on the WCS system's event bus
- Implemented improvements in the WCS system's database, using Entity Framework/C#, to allow for many to many relations between destinations and routing points, saving space and preventing the duplication of data in the database
- Wrote and published company wide documentation detailing the functionality of each aspect of the WCS system (printer management, event bus testing, routing and sortation of decision points, PLC communications, system configuration)
- Oversaw, tested, and debugged packaging systems for the Dell Syncreon Project; this project required coordination with the mechanical and controls engineering team to make sure each step in the package evaluation process went as expected
- Oversaw, programmed, and presented a robotics system for UPS involving a 10 AGV system that organized and distributed small sized packages coming from a conveyor to a desired tipper location; the project was essentially a state machine, requiring coordination between all AGVs and the conveyor system to ensure that filled totes were transported to their correct destination

Infosys Huntington Beach, CA

Software Engineer

October 2020 - February 2021

Goldman Sachs's Project

- Designed and programmed an improved system based on Kafka to increase efficiency of contract handling, retrieval, and dispersion
- The Kafka system was implemented in Java and was used for tracking financial transactions within the Goldman Sachs' system
- Designed and implemented improvements to the Kafka systems (in Java) (i.e. ensuring that the correct retry exceptions are used and handled correctly)
- Conducted root cause analysis for recurrent system failures, proposed solutions to technical leads, and implemented code/system changes to reduce the number of failures per day
- Conducted technical support for user queries and alerts in order to provide solutions to help operations continue smoothly
- Maintained and regulated GCP compute instances through traffic analysis, health analysis, and Load Balancers
- Completed Infosys's Purdue Cybersecurity Essentials Training which emphasized endpoint management, incident analysis, exploitation methods, information encryption methods, web application vulnerability recognition, malware classification, firewalls, and DMZ

Oak Analytics

Agoura Hills, CA

Software-Engineer Intern

June 2018 - August 2018

- Designed and built a new testing application with Android Studio, providing the company with insight into the types and frequency of information lost during the Bluetooth/scanner transfer process; application included performance graphs and data tables

Interests

Music production, audio engineering, microcontroller wiring and programming, basketball

